Digital Twin of sorting Conveyor Belt in CoopliaSim

Course Project (SM2007) Report

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Abstract

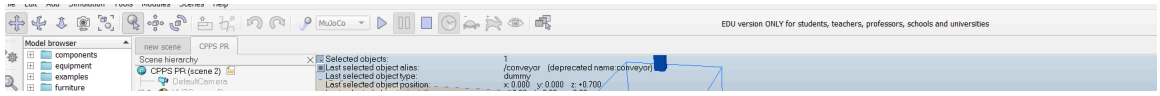
This project implements a digital twin of a conveyor-based replenishment system, utilizing the foundational logic of the **e-Kanban** framework—a digitized evolution of the landmark production system developed by **Toyota**. By integrating a probabilistic **(s,S) inventory policy**, the **simulation(*graph 1*) demonstrates** a responsive 'pull' mechanism where material flow is triggered by real-time demand rather than static forecasting.

A **Digital Twin of a sorting conveyor** is the critical "bridge" for an e-Kanban system because it solves the biggest problem in traditional manufacturing: **The Visibility Gap.**

In a standard factory, the "Physical" (the boxes on the belt) and the "Digital" (the inventory database) are often disconnected. **A Digital Twin acts as the live sensor layer that makes e-Kanban possible.It can capture the number of objects leaving the belt, bin where sorted objects go being full or not.**

This is crucial to know if we can replenish back to "S" inventory, removing the need of manual logging.

It also provides remote accessibility for monitoring and controlling the belt. **In this project the belt can be controlled from CoppliaSim 's user interface**. Thus making seamless link between digital belt and physical belt

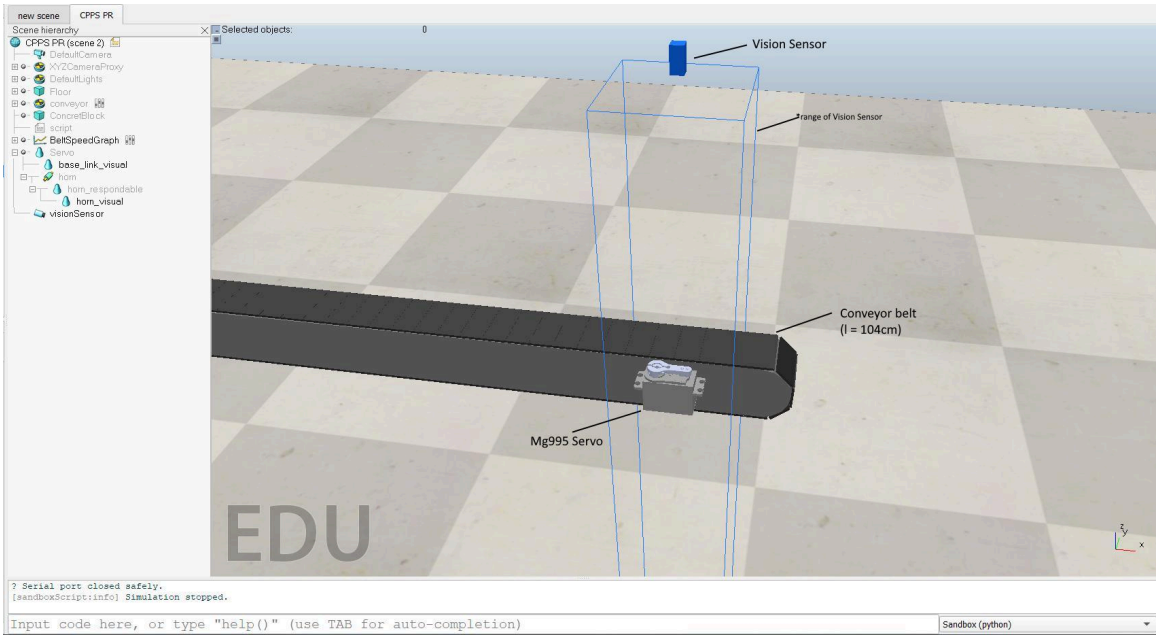


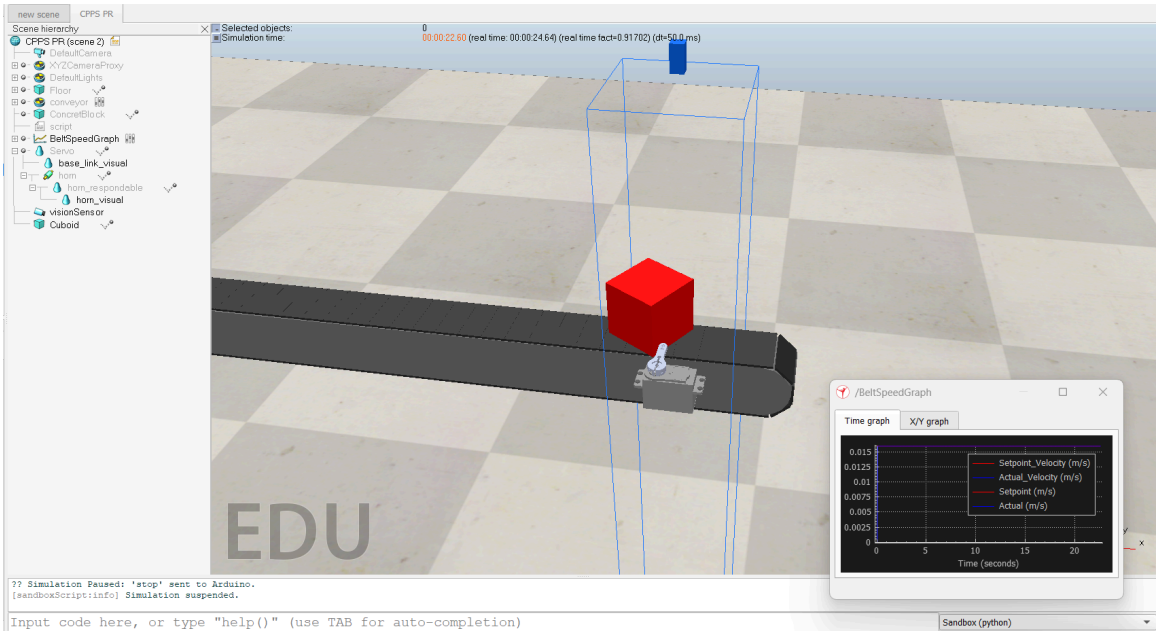
[fig. A]

[*User Interface to start, pause, end sim in CoppliaSim to control physical Belt*]

[https://drive.google.com/file/d/14XclNj89roTvcz4Pm-0UrVjVHa1qFQUF/view?usp=sharing]

Graphical Abstract

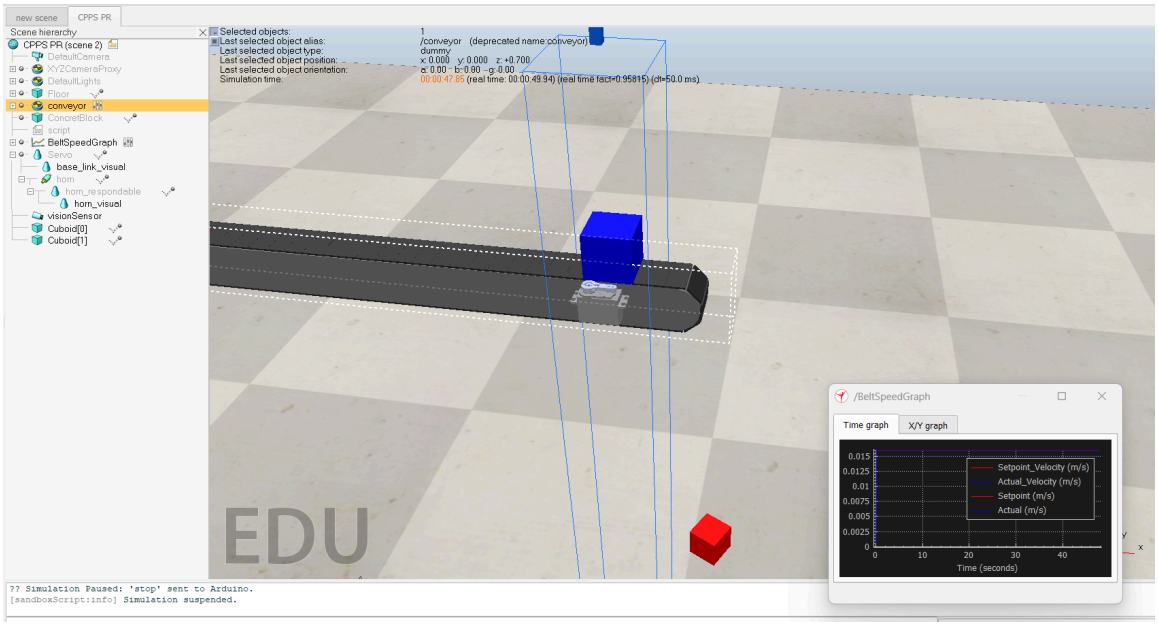




[fig. B(Top)] & [fig. C(Bottom)]

[Real life action video from decision taken by Digital Twin]

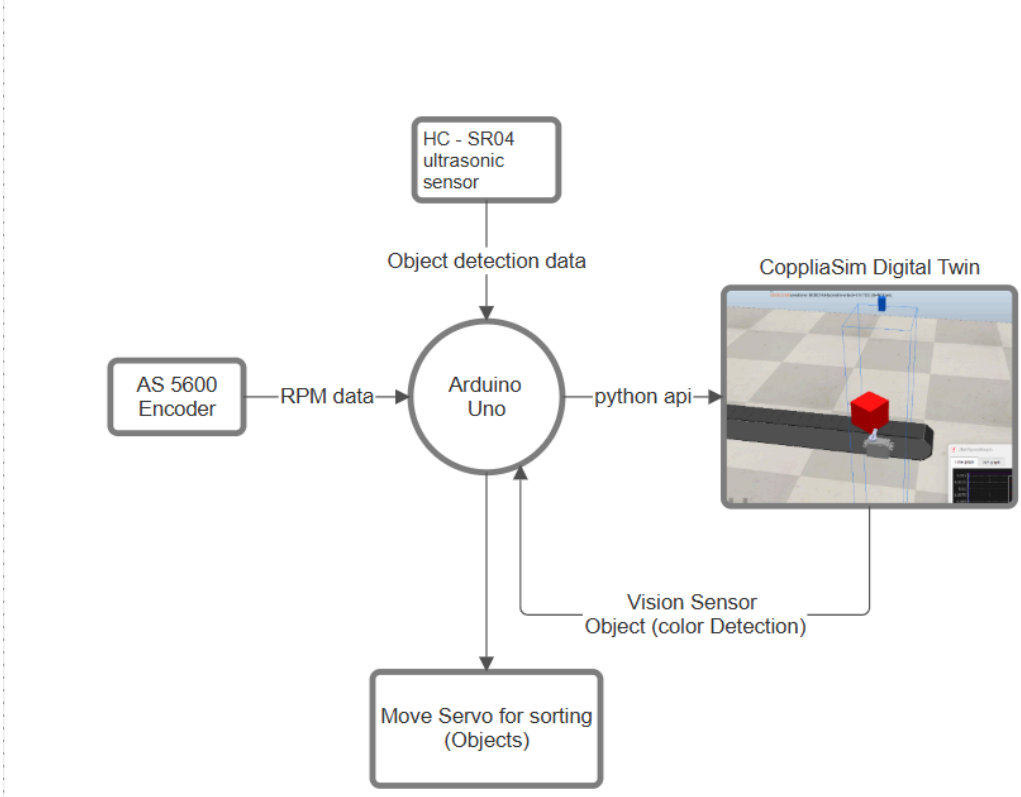
[https://drive.google.com/file/d/1-J29LP1_3ZOTmwoy6cZEDh1Hc09_1wrU/view?usp=sharing]



[fig. D]

*[*CoppliaSim Scene for Digital Twin of Sorting conveyor belt]*

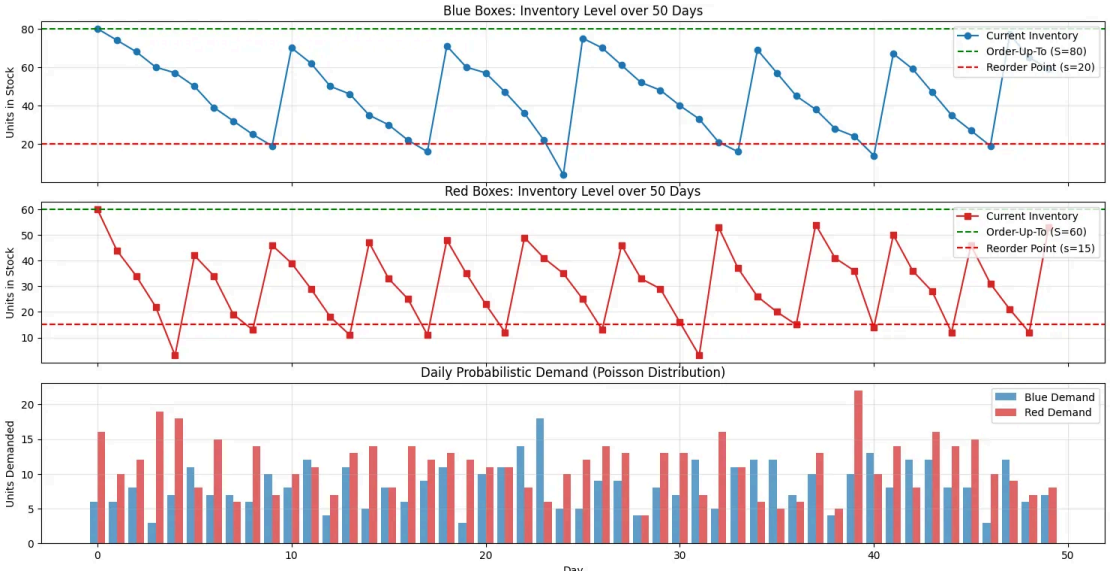
[https://drive.google.com/file/d/1YAIo5bQK0SgARg5g1IUKcTgfTDlekovy/view?usp=sharing]



[fig. E]

[Flowchart for Digital Twin of Sorting conveyor belt]

*[*it can be seen that servo has turned by 90 degrees when red block is detected]*



[graph 1]

*[python simulation of Red and Blue boxes inventory with **sS inventory Policy** with **probabilistic demand over the period of 50 days**]*

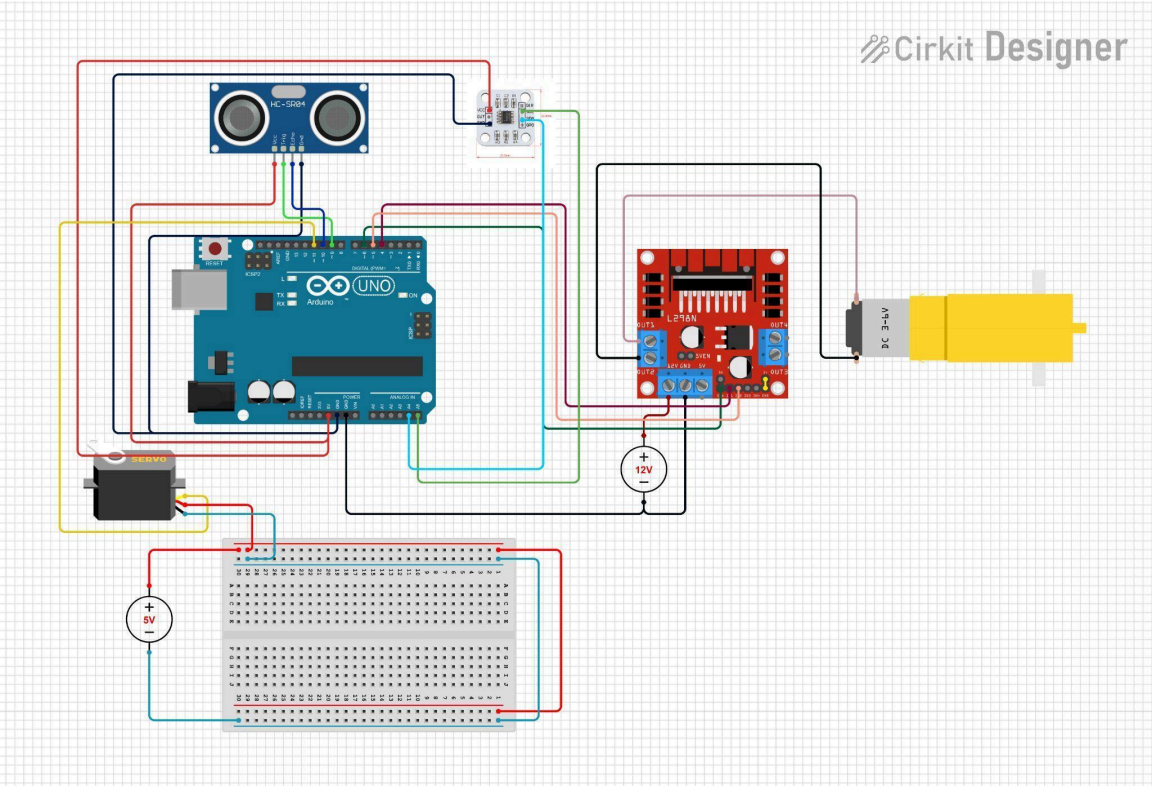
1. Introduction

In the era of Industry 4.0, Digital Twins serve as a cornerstone for optimizing manufacturing logistics and operational efficiency. This project addresses the challenge of real-time synchronization between physical conveyor systems and their virtual counterparts. While many simulations operate in isolation, this project creates a unified environment where physical dynamics (velocity and dimensions) are mirrored in a 3D digital workspace.

The primary objective was to create a 'one-of-a-kind' system where the CoppeliaSim UI dictates the physical state of the machine (Start/Pause/Stop) while the physical hardware provides real-time motion telemetry back to the twin.

2. Schematic Diagram

https://app.cirkitdesigner.com/project/d9228d28-0b95-4da0-bed4-be7adbecac25



[*fig. F*]

[Schematic Diagram:UNO R3 clone, AS5600, Motor Driver, MG995, HC SR04]

- Connection Details:
- AS5600 Encoder: Connected via I2C (SDA/SCL) to the microcontroller(UNO R3).
- MG995 Servo:Connected to PWM capable pin of MCU(~11).

- HC SR04 : Ultrasonic sensor for object detection(`trigPin` → Pin 9 `echoPin` → Pin 10).
- Main Drive Motor(L298N):Powers and controls direction of belt.

3. Methodology and Working Principle

3.1. Velocity and Dimension Calibration

To ensure high fidelity, the velocity of the digital twin's belt is mapped 1:1 to the IRL belt using the AS5600 absolute magnetic encoder. A custom software filter was developed to convert absolute angle readings to incremental angle readings.Special calibration function is also defined in arduino code for calculating speed of belt.

3.2. Custom URDF Integration

The MG995 Servo and 25T Aluminum Horn were imported as a Custom URDF into CoppeliaSim to ensure collision physics and rotational limits accurately reflect the physical hardware. We used SolidWorks ROS to URDF importer to convert our model of MG995 servo motor to be controllable(i.e., to move servo horn) in CoppliaSim.

3.3. Vision Sensor & Sorting Logic

Virtual Vision Sensor(CoppliaSim) takes the decision for Digital and Physical Belt, it works due to both of them being synchronous. The constant velocity measurement from the motor plays a crucial role in both of them being synchronous.

Red Block: Vision sensor triggers a 90-degree rotation.

Blue Block: Vision sensor triggers return to home (0-degree) position.

3.4. Bidirectional Control Loop

The system uses the CoppeliaSim Python API to link the virtual state (Start/Pause/Stop) directly to the physical motor.

3.5. Ultrasonic Sensor

The ultrasonic sensor on the Physical belt is responsible for taking the action of spawning that particular component [Red box, Blue box] on the digital twin. Serial command sent by Arduino is interpreted by the python api to spawn box.

4. List of Items/Components Used

Category	Component	Quantity
Microcontroller	Arduino Uno R3 (*clone)	1
Sensors	AS5600 Magnetic Absolute Encoder	1
	HC-SR04 Ultrasonic Sensor	1
Actuators	MG995 High Torque Metal Gear Servo	1
Mechanical	25T Aluminum Servo Horn	1
Electrical	5V breadboard power supply	1
Drive	Conveyor Drive Motor	1
Software	CoppeliaSim, Python 3.x, Python API	-

5. Results and Discussion

The implementation demonstrated that the Digital Twin could maintain synchronization within a negligible error margin. The responsiveness of the virtual vision sensor-driven sorting mechanism for the physical and digital world(conveyor belt) proved robust, with minimal delay between virtual detection and physical action. There was no vision or proximity sensor near the physical sorting Servo motor, position of boxes was estimated through digital Twin only. This shows negligible error margin in Digital Twin application

6. Conclusion

This project establishes a functional framework for a Real-Time Digital Twin for sorting of objects. This system highlights the potential of low-cost components when paired with sophisticated software integration for modern manufacturing.

Videos for in real life application of Twin for those specific cases are provided in Graphical Abstract.This clip where extracted from complete video, to get precise information faster.

Complete video of digitalTwin and simulation
https://drive.google.com/file/d/1LI32QK99ggMQIcYXvtwZXc7V4zrX4eUZ/view?usp=sharing

Digital Twin video[screen recorded version]

https://drive.google.com/file/d/1-UraNXet-4LJXo6RJURt3YZgzJIG9eDS/view?usp=sharing

7. References

1. [1] Coppelia Robotics. 'CoppeliaSim Python API Documentation.'
2. [2] AS5600 Datasheet. '12-Bit Programmable Contactless Potentiometer.'
3. [3] IIITDM Jabalpur. 'Smart Manufacturing Coursework(SM2007).'
4. [4] CoppeliaSim Forum. Technical discussions.